

Predicting Zero-Shot Success from Representation Geometry

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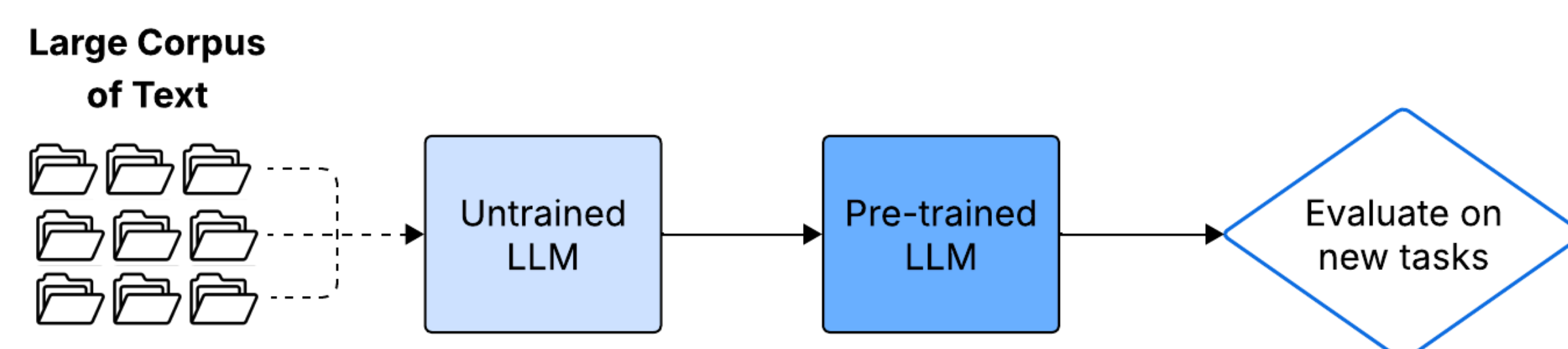
Motivation

Evaluating large language models on new tasks requires expensive brute-force benchmarking. Zero-shot performance is highly unpredictable, with no reliable way to anticipate success or failure on superficially similar tasks. This raises the question of whether pretrained representation geometry captures task structure related to generalization difficulty, enabling lightweight prediction of downstream performance without full task-specific training.

Background

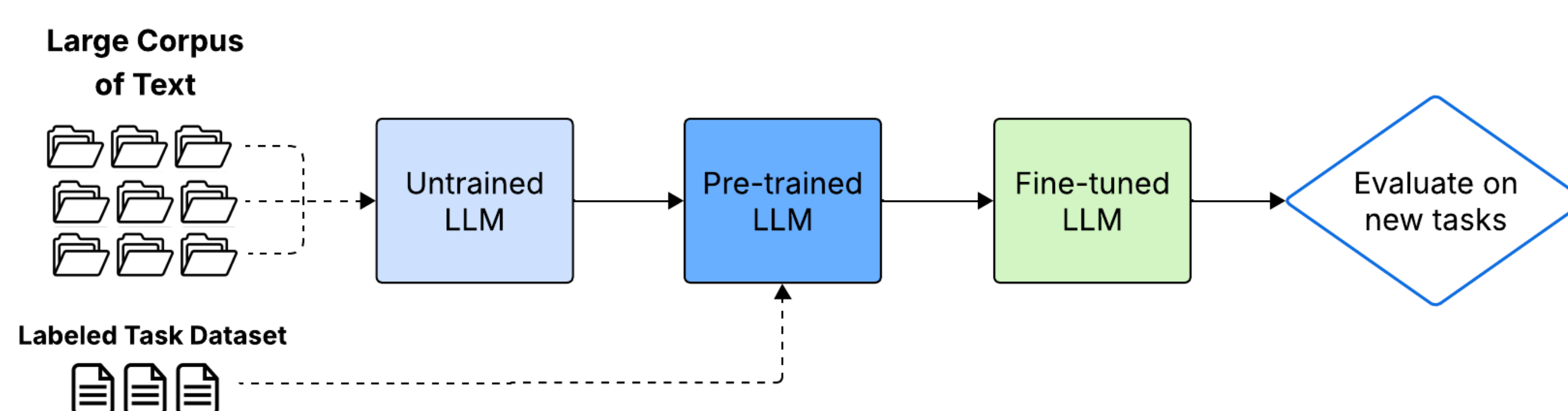
Zero-shot

- Evaluating a pretrained model on a new task without any task-specific training.



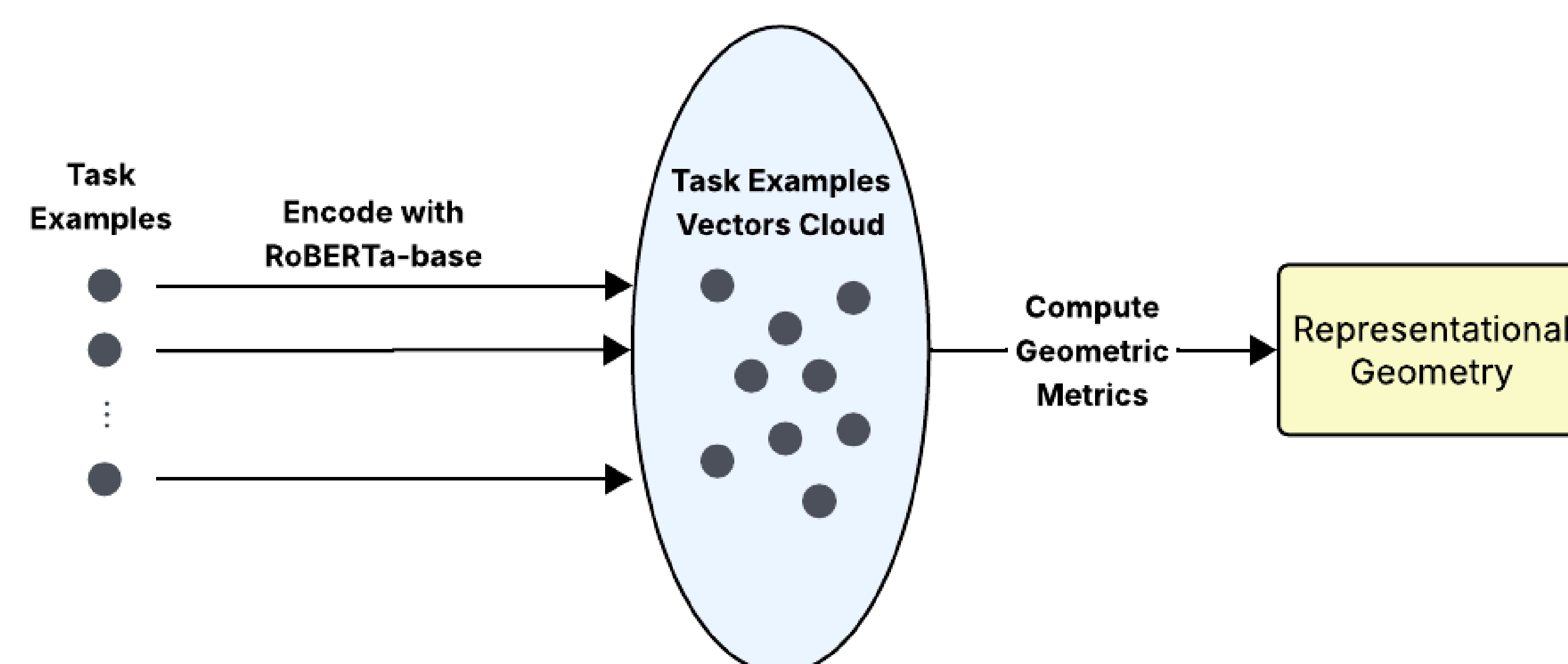
Fine-tuning

- Adapting a pretrained model to a specific task using labeled data to improve performance.



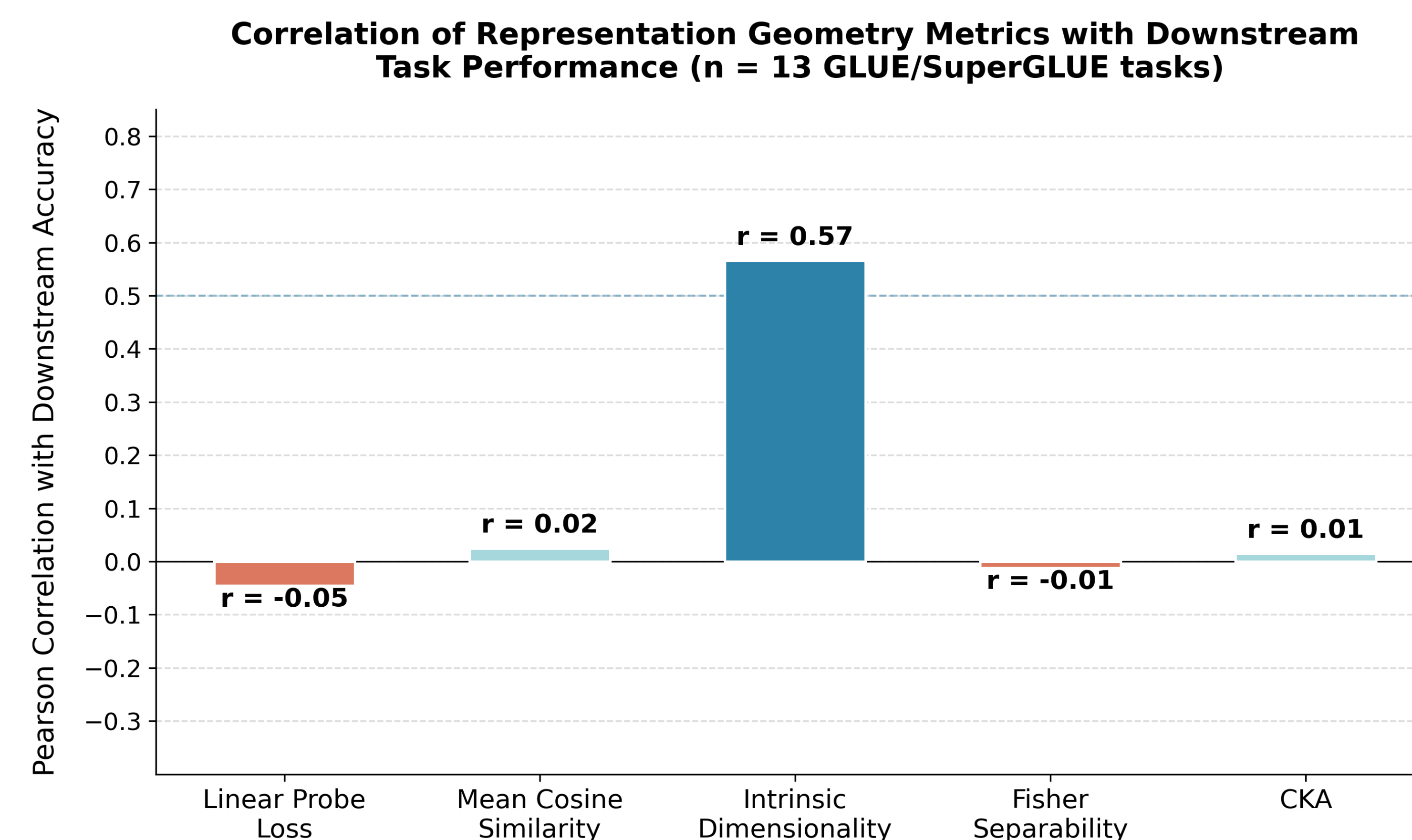
Representational Geometry

- Each task's examples are encoded by RoBERTa into points in a high-dimensional (768-dimensional) embedding space.
- The geometric structure of these embeddings may reflect how easily the task can be learned.

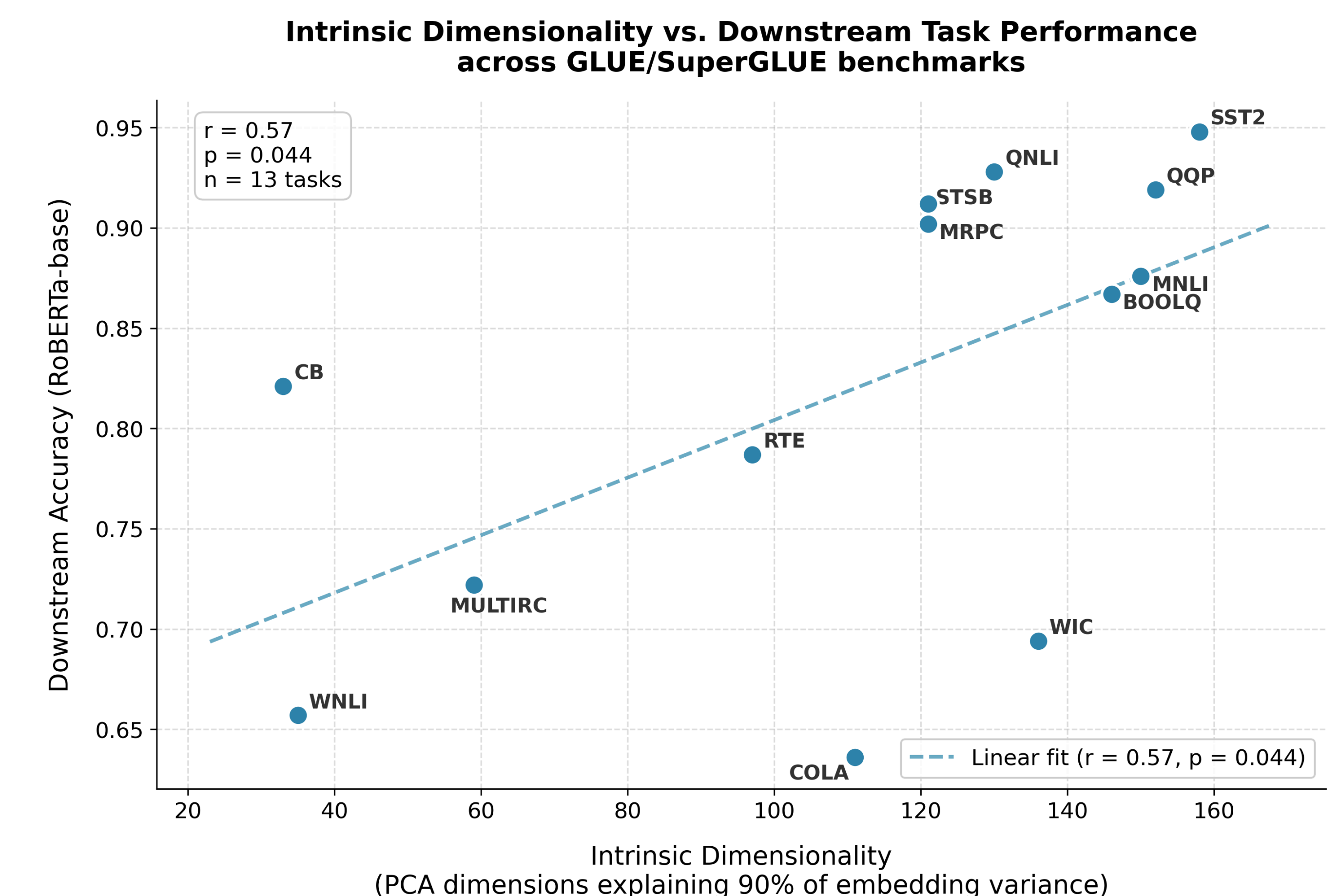


Results

Four of five metrics showed weak, non-significant correlations with downstream accuracy ($|r| < 0.20$). Intrinsic dimensionality was the sole exception, exhibiting a moderate, statistically significant positive correlation ($r = 0.57$, $p = 0.044$, $n = 13$). This suggests that tasks whose embeddings occupy higher-dimensional subspaces in the pretrained representation space tend to be associated with higher downstream performance.



Note that the observed correlation ($r = 0.57$, $p = 0.044$) should be interpreted cautiously given the small sample size ($n = 13$).



Discussion

- Most geometric metrics computed on pretrained embeddings fail to predict downstream task performance, likely because fine-tuning substantially reshapes higher-layer representations in task-dependent ways (Merchant et al., 2020), decoupling pretrained structure from final task-specific geometry.
- All metrics are computed in the pretrained representation space, which is also used directly in zero-shot inference. We evaluate against supervised accuracy as a proxy for task difficulty, testing whether pretrained structure contains signals about task learnability.
- Across 13 benchmarks, most metrics show weak, non-significant correlations with performance. Intrinsic dimensionality is the exception, showing a moderate, statistically significant correlation, suggesting it captures a more global property of how task examples occupy the representation space.
- This aligns with Aghajanyan et al. (2021), who show tasks differ in intrinsic adaptation complexity in low-dimensional parameter subspaces. Our results suggest a complementary signal in representation space: tasks spanning higher-dimensional regions may reflect greater structural complexity and be more learnable.

Future Directions

- Track geometry pre- and post-fine-tuning to evaluate how intrinsic dimensionality changes under task-specific adaptation.
- Test whether pretrained intrinsic dimensionality predicts true zero-shot performance beyond supervised proxies.
- Scale to more tasks and architectures to assess robustness and generality of the observed relationship.

References

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